

Branched Propylene Derivatives by Radical-Mediated Grafting of an Acrylate-Functionalized Nitroxyl

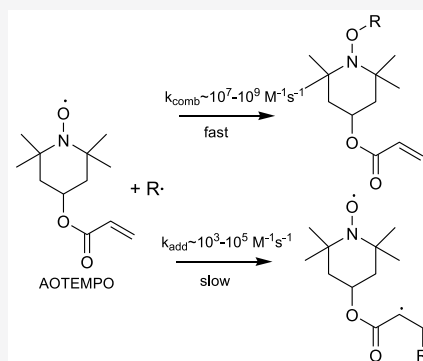
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ABSTRACT: Peroxide-initiated radical chemistry for producing a long-chain branch (LCB) architecture to linear, isotactic propylene homopolymer is described, wherein degradation of the polymer matrix is mitigated by a nitroxyl bearing a polymerizable functional group. Trapping of alkyl macroradicals by nitroxyl yields a macromonomer derivative that crosslinks when exposed to residual initiator-derived radicals. As such, formulations containing 4-acryloyloxy-2,2,6,6-tetramethylpiperidine-1-oxyl (AOTEMPO) and peroxide transform linear polypropylene (PP) into gel-free derivatives (LCB-PP) with predictable melt-state rheological properties as well as robust strain-hardening characteristics. These derivatives demonstrate enhanced nucleation during crystallization from the melt, leading to higher crystallization temperatures that are characteristic of LCB polyolefins.



1. INTRODUCTION

Polypropylene (PP) is a commodity thermoplastic that is valued for its strength, toughness, and chemical stability. Unfortunately, commercialized propylene polymerization catalysts do not provide long-chain branched (LCB) architectures.^{1,2} Rather, they yield linear materials that are ill-suited to melt-state processing operations that involve extensional deformations, such as thermoforming, foaming, and film blowing.³ Therefore, there is continued interest in postpolymerization modifications of linear PP that provide gel-free LCB-PP derivatives with uniform branching and molecular weight distributions.

The saturated hydrocarbon structure of PP limits its modification chemistry to radical-mediated reactions initiated by peroxide thermolysis, photoinitiator activation, mechanical shear,⁴ or high energy irradiation.^{5,6} We are concerned with high temperature, melt-state chemistry that can be carried out in conventional polymer processing equipment. These operations are complicated by the susceptibility of tertiary alkyl macroradical intermediates to β -scission,⁷ which can lead to extensive losses of melt viscosity.⁸ The challenge in devising chemistry to produce LCB-PP is to overcome this degradation process, such that the requisite branch density is achieved while preserving matrix molecular weight.⁹

One approach involves the grafting of multifunctional monomers such as triallyl trimesate (TAM) and trimethylolpropane triacrylate (TMPTA), which build branching and molecular weight by repeated grafting of C=C functionality to the polymer backbone. These processes are plagued by low grafting efficiency and extensive monomer oligomerization,

and they yield bimodal product distributions comprised of a small population of hyperbranched chains that are dispersed in a degraded linear matrix.¹⁰ Alternative two-step branching strategies reduce the extent of this bimodality through a sequence of monofunctional monomer grafting followed by crosslinking by nonradical chemistry.¹¹ Examples include grafting of vinyltriethoxysilane (VTES) then moisture curing the PP derivative, and grafting of maleic anhydride¹² followed by crosslinking using imidation reactions of multifunctional amines. While more effective than coagent-based products, these LCB-PP materials exhibit melt-state viscosities that fall well below their parent materials, thereby complicating the task of formulation developers.

We have shown that functionalized nitroxyls can be used to prepare thermoset derivatives of PP homopolymers.¹³ Polyolefin grafting is achieved by quenching alkyl macroradicals by combination with nitroxyl to produce alkoxyamine intermediates bearing a polymerizable functional group (Scheme 1). Since radical trapping operates close to the diffusion limit of reaction velocity,¹⁴ macroradical lifetimes are reduced to the point where PP degradation is eliminated entirely.^{15,16} Upon complete conversion of nitroxyl functionality to alkoxyamine, oligomerization of polymer-bound C=C

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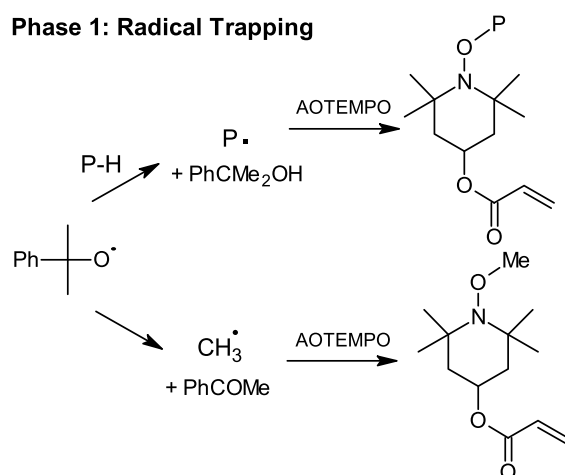
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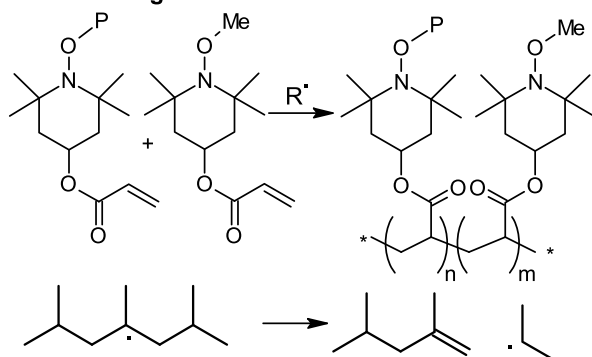


Scheme 1. Mechanism AOTEMPO-Mediated Synthesis of LCB-PP

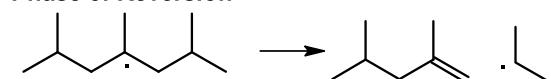
Phase 1: Radical Trapping



Phase 2: Oligomerization and Scission



Phase 3: Reversion



functionality by residual initiator establishes the desired covalent network of polymer chain segments.¹⁷ Given sufficient peroxide and a suitable amount of functional nitroxyl, thermoset PP products with gel contents in excess of 90% have been achieved using a simple compression molding process.¹³

Our goal was to adapt functional nitroxyl chemistry to generate gel-free LCB-PP derivatives that provide exceptional strain-hardening characteristics and high polymer matrix viscosity. This report begins with comparisons of LCB-PP formulations containing acrylated TEMPO (AOTEMPO) relative to benchmarks prepared by vinyl trialkoxysilane grafting + moisture curing. These comparisons include careful measurements of oscillatory shear rheology and extensional rheology properties of LCB-PP melts to probe polymer chain architecture. This is followed by detailed studies of the influence of LCB structures on melt-state crystallization dynamics and product morphology under isothermal and nonisothermal conditions.

2. EXPERIMENTAL METHODS

2.1. Materials. Acryloyl chloride (>97%), butylated hydroxytoluene (BHT, >99%), dicumyl peroxide (DCP, 99%), 4-hydroxy-2,2,6,6-tetramethylpiperdin-1-oxyl (HOTEMPO, 97%), isotactic polypropylene (PP, $M_w = 340\,000$ g/mol, $M_n = 97\,000$ g/mol), 2,2,6,6-tetramethylpiperdin-1-oxyl

(TEMPO, 98%), pyridine (>99.9%), and triethylamine (98%) vinyltriethoxysilane (VTES, 97%) were purchased from Sigma-Aldrich. 2,5-Bis(tert-butylperoxy)-2,5-dimethyl-3-hexyne (L-130, 56%) was purchased from Elf Atochem, while dibutyl tin dilaurate (DBTDL, >94%) was purchased from Alfa Aesar. All chemicals were used as received.

2.2. AOTEMPO Synthesis. AOTEMPO was prepared as described previously¹⁸ with minor modifications. A solution of triethylamine (4.30 mL, 3.12 g, 30.9 mmol) in dry benzene (14 mL) was added dropwise to a stirred solution of HOTEMPO (1.36 g, 7.90 mmol) in dry benzene (20 mL) at room temperature under N_2 . A solution of acryloyl chloride (675 μ L, 0.379 g, 4.18 mmol) in dry benzene (14 mL) was added dropwise at room temperature and the solution stirred for 24 h, after which a second dropwise addition of acryloyl chloride (340 μ L, 0.379 g, 4.18 mmol) in benzene (7 mL) was performed. The reaction was stirred for an additional hour before removing solids by vacuum filtration and washing with benzene. The solvent was removed from the filtrate under vacuum, yielding orange crystals that were recrystallized from cyclohexene. Yield: 76%; mp 96–97 °C; and lit 102–104 °C.¹⁸

To facilitate characterization of the paramagnetic molecule, AOTEMPO was reduced to its *N*-hydroxyl amine with phenylhydrazine in an NMR tube. ¹H nuclear magnetic resonance (NMR) (400 MHz, $CDCl_3$, δ , ppm): 6.29 (1H, dd, $J_1 = 2$ Hz, $J_{2(trans)} = 17$ Hz), 5.99 (1H, dd, $J_{2(trans)} = 17$ Hz, $J_{2(cis)} = 11$ Hz), 5.72 (1H, dd, $J_1 = 2$ Hz, $J_{2(cis)} = 11$ Hz), 5.05 (1H, tt, $J_{2(trans)} = 11$ Hz, $J_{2(cis)} = 4$ Hz), 1.89 (2H, dd, $J_1 = 12$ Hz, $J_{2(cis)} = 4$ Hz), 1.65 (2H, t(broad), $J_1 = 12$ Hz, $J_{2(trans)} = 11$ Hz), 1.19 (6H, s), and 1.16 (6H, s).

2.3. LCB-PP Synthesis. Polypropylene (5 g) was ground and divided evenly between three 60 mL plastic bottles. Desired reagent loadings for each formulation were dissolved in acetone (1.5 mL) and the solution was distributed evenly between the polymer powders within the bottles, tumble-mixed intermittently until the acetone had evaporated (2–3 h).

An Atlas Laboratory Mixing Molder heated to 180 °C was charged with the contents of one bottle and the compound was mixed for 18 min (5 half-lives of the peroxide). The product was then removed from the mixer, and the process was repeated for each subsequent bottle. Products from different bottles in the same batch were ground and combined.

Peroxide-initiated graft modification of PP with vinyltriethoxysilane was carried out as described above. Combined PP-g-VTEOS samples were solution-coated with 5 μ L/g DBTDL and 5 wt % Irganox-1010 and mixed using the Atlas Laboratory Mixing Molder for 3 min. The resulting stabilized PP-g-VTEOS samples were pressed into films approximately 1 mm thick using a Wabash press, then moisture-cured by submerging in boiling water for 2 h.

2.4. Rheological Characterization. Compression-molded discs 25 mm in diameter and 1.0–1.4 mm in thickness were prepared with a heated hydraulic press. Storage modulus (G'), loss modulus (G''), and complex viscosity (η^*) were measured using an Anton Paar MCR-301 rheometer equipped with 25 mm diameter parallel plates using a gap of 1 mm. Samples were measured over a frequency range of 0.05–100 rad/s at a constant strain of 3% and a temperature of 180 °C. Strain sweeps were used to confirm that measurements were acquired in the polymer's linear viscoelastic region. Uniaxial extensional rheometry was performed with a universal testing platform (Anton Paar SER-2). Sample films were cut to a width of 10

mm, and testing was performed at 180 °C using strain rates of 0.1, 1, and 10 s⁻¹.

2.5. Gel Content. Measurements were made in accordance with ASTM D2765-16.¹⁹ Weighed samples were enclosed in 120 mesh stainless steel cloth and extracted with a xylene solution containing 1 wt % BHT under reflux for 12 h. Samples were then dried to constant weight, and gel content was reported as the mass fraction remaining of the original sample.

2.6. Differential Scanning Calorimetry. Measurements were acquired using a TA Instruments DSC Q100 using hermetically sealed aluminum pans. All samples were held for 3 min at 200 °C to remove the thermal history. For nonisothermal testing, samples were cooled from 200 to -40 °C at a rate of 5 °C/min, then held at -40 °C for 5 min before heating the sample back to 200 °C at the same rate. For isothermal testing, samples were cooled to the target temperature (130 or 140 °C) at a rate of 30 °C/min before maintaining the target for 1 h.

2.7. Hot Stage Microscopy. Isothermal crystallization experiments were performed using a Linkam CSS 450 hot stage mounted on an Olympus BX51 optical microscope. Samples were heated to 200 °C and pressed into a thin film by reducing the plate gap to 50 μm, and the temperature was held at 200 °C for 5 min to eliminate any thermal history. The film was then cooled to 140 °C at a rate of 30 °C/min and the temperature was kept constant for 30 min. Images were captured using a Sony ExwaveHAD 3 CCD digital recorder.

3. RESULTS AND DISCUSSION

3.1. Trapping Ratio Screening. Radical-mediated polyolefin modifications involving functionalized nitroxyls progress through three distinct phases: (1) an induction period where alkyl radicals are trapped by nitroxyl to produce alkoxyamines, (2) an oligomerization stage where acrylate functionality is activated to produce a covalent network, and (3) a stage where conventional polymer modification occurs unencumbered.^{20,21} In the case of PP, the third stage involves degradation arising from the β-scission of tertiary macroradicals.¹³ The duration and yield of each phase is dictated by the initiator decomposition rate as well as the relative amount of nitroxyl versus peroxide initiator, as described by a formulation's trapping ratio (TR)¹⁷

$$TR = \frac{[\text{nitroxyl}]}{\# \text{ radicals/peroxide} \times [\text{peroxide}]}$$

Preliminary experiments were conducted to identify appropriate reagent loadings for preparing LCB-PP from the linear PP starting material. A fixed peroxide loading was used in conjunction with different AOTEMPO concentrations, resulting in trapping ratios that varied from 0 to 1. Figure 1 illustrates the oscillatory shear rheology data measured for the resulting LCB-PP derivatives.

The PP parent material produced rheological data that is entirely consistent with a linear chain architecture (Figure 1). The complex viscosity approached a Newtonian plateau at low oscillation frequencies, where sufficient time allowed for extensive chain segment relaxation, as evidenced from the approach of G' toward a terminal flow condition. Pronounced shear thinning was observed, with the complex viscosity dropping from a maximum of 11 400 Pa·s at 0.05 rad/s to a low of 704 Pa·s at 100 rad/s. Treatment of this PP with

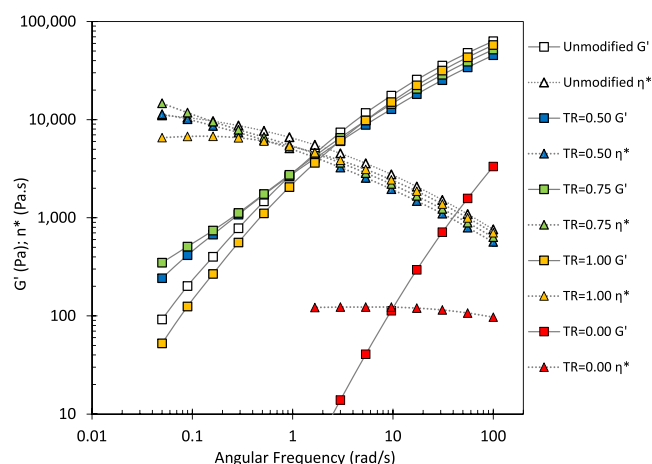


Figure 1. Storage modulus and complex viscosity of LCB-PP prepared with varying AOTEMPO trapping ratios ([L130] = 8 μmol/g; [AOTEMPO] = 0–32 μmol/g).

peroxide alone (TR = 0) resulted in severe degradation, as evidenced by orders of magnitude reductions of η^* and G' .

Including AOTEMPO in the peroxide-initiated PP formulations mitigated chain scission substantially, as indicated by the preservation of rheological properties in the high-frequency domain. Melt-state responses over these timescales are dominated by small segment dynamics [1], making η^* and G' data highly sensitive to the molecular weight of the polymer matrix. Complex viscosities measured at 100 s⁻¹ declined in the order of unmodified PP $\approx$ TR = 1.00 > TR = 0.75 > TR = 0.50 $\gg$ TR = 0.00 (Figure 1). Note that TR = 1.00 provides sufficient nitroxyl to quench all alkyl radicals generated by initiator thermolysis. Therefore, this formulation progresses only through an induction phase of an AOTEMPO-mediated process, yielding pendant alkoxyamine functionality whose crosslinking potential is not realized due to the lack of residual initiator.¹³ As such, maintenance of molecular weight and, by extension, melt viscosity is expected.

Formulations with intermediate trapping ratios progress through all three phases: induction, oligomerization, and unfettered reactivity. Using a TR of 0.75 quenches 75% of the initiator in phase 1, allowing 25% of the L-130 in the material to oligomerize acrylate functionality (phase 2) and degrade the PP matrix (phase 3). Lowering the AOTEMPO loading to a TR of 0.50 traps 50% of the initiator during an abbreviated induction phase, yielding a smaller amount of macromonomer functionality and leaving more residual L-130 to initiate subsequent acrylate oligomerization and PP degradation.²⁰ Hence, the observed order in high-frequency η^* data is consistent with expectations based on the prevalence of phase 3, wherein PP degradation by tertiary macroradical cleavage is the dominant reaction that affects molecular weight.

Rheological properties at low oscillation frequencies are dominated by chain entanglement effects, with LCB architectures producing higher G' and η^* values compared to linear polymers of the same molecular weight [1]. The data plotted in Figure 1 show elevated low-frequency properties for the TR = 0.50 and TR = 0.75 formulations compared to the parent material. Storage modulus values for these samples did not exhibit terminal flow over the accessible low-frequency domain, owing to the slow relaxation of chain entanglements within an LCB-PP network. In contrast, the TR = 1.00 sample showed no evidence of branching, consistent with the stunting

of acrylate oligomerization by the insufficient initiator. Based on these results, a trapping ratio of 0.75 was selected for further research into the ability of AOTEMPO formulations to produce high-molecular-weight, gel-free LCB-PP with extensional strain-hardening characteristics.

3.2. Comparative Analysis of LCB-PP Formulations.

We have previously examined LCB-PP chemistry involving single-step grafting of multifunctional acrylic and allylic coagents, as well as approaches involving monomer grafting followed by derivatization of polymer-bound functionality.¹¹ A sequence of radical-mediated addition of vinyltriethoxysilane to PP followed by moisture curing to produce LCB-PP was adopted as a reference for the present study, as it provides more uniform branching and molecular weight distributions than coagent-based techniques.¹¹ Three AOTEMPO formulations and two VTES formulations were prepared as detailed in Table 1. Although the same polymer, temperature, and peroxide were used in both series, different reagents were applied in recognition of fundamental differences in underlying chemistry.

Table 1. LCB-PP Formulations and Product Properties

sample	[L130] ($\mu\text{mol/g}_{\text{PP}}$)	[AOTEMPO] ($\mu\text{mol/g}_{\text{PP}}$)	[VTES] ($\mu\text{mol/g}_{\text{PP}}$)	LCB-PP gel content ^a
PP-AO1	6.0	18		<0.5%
PP-AO2	8.0	24		<0.5%
PP-AO3	9.7	29		5%
PP-S1	1.5		263 (3 wt %)	<0.5%
PP-S2	1.5		350 (4 wt %)	2%

^aWeight percent of xylenes-insoluble material.

Studies of conventional LCB-PP formulations have shown that, where extensional strain hardening is concerned, uniform branching distributions are not an absolute requirement. This is fortunate, given the statistical nature of chain activation by peroxide-initiated graft modifications. Given that H-atom transfer from each mer of PP is equally probable, and that abstraction events are independent, the macroradical population generated by a peroxide will be distributed, such that an average of one abstraction per molecule is spread among chains having 0, 1, 2, 3, 4+ activations. This nonuniformity of PP activation plagues both AOTEMPO and VTES LCB-PP syntheses and should be recognized as a fundamental limitation. In general, attempts to increase chain activation yields are met with an increase in multiply functionalized chains, which produce a gel fraction in the branched product when the macromonomer functionality is converted to crosslinks.

The challenge in developing an effective VTES-based branching formulation is coping with the monomer's inability to quench the scission of tertiary alkyl macroradicals. At practical VTES loadings, the rate of C=C addition to PP macroradicals is insufficient to prevent concurrent degradation. As a result, silane-grafted derivatives have low molecular weight and a predominance of unmodified chains. Since moisture curing establishes a covalent network only from those chains that bear silane functionality, the branched product typically contains an LCB population that is dispersed in a matrix of small linear chains. The VTES formulations listed in Table 1 provided a gel-free sample (PP-S1) and a sample that

contained 2 wt % of xylenes-insoluble polymer (PP-S2). The former is more desirable from a polymer processing perspective, while the latter is more typical of LCB-PP materials generated by this synthetic method.

When compared to AOTEMPO formulations, the VTES trials used a small amount of peroxide initiator and a much greater amount of monomer. Minimizing the L-130 concentration was required to limit the extent of molecular weight degradation during the PP functionalization step of the branching process. Relatively high VTES loadings were needed to encourage functionalized chain production by providing more monomer than the number of PP molecules in the formulation. Note that increased VTES availability is proven to improve silane graft contents, but it has a marked tendency to introduce grafts in close proximity on a single chain, due to a favorable intramolecular 1,5-hydrogen atom abstraction in the grafting sequence.²²

The AOTEMPO formulations listed in Table 1 maintain a trapping ratio of 0.75 while employing relatively high initiator loadings and small monomer-functionalized nitroxyl concentrations. The tolerance of this synthetic approach to peroxide concentration stems from the efficiency of alkyl radical trapping by nitroxyl to produce pendent acrylate macromonomer groups at the site of H-atom abstraction. This rapid alkyl radical quenching is a stoichiometric process with no kinetic chain length. Therefore, achieving the requisite acrylate content requires the use of more peroxide than a comparable VTES formulation. Note that the extent of PP modification provided by the amounts of L-130 and AOTEMPO is a function of the H-atom abstraction efficiency of initiator-derived alkoxy radicals. At the temperatures applied to melt-state polyolefin modifications, the fragmentation of alkoxy radicals to $\text{CH}_3\cdot$ + ketone can be substantial. For example, dicumyl peroxide thermolysis to cumyloxyl at 160 °C provides a macroradical yield of just 35% for PP.⁷ In a functional nitroxyl formulation, this abstraction efficiency would convert only 35% of AOTEMPO to PP-bound acrylate, with the remainder being methyl alkoxyamine. The PP-AO formulations described herein provided the requisite degree of PP modification, yielding two gel-free samples (PP-AO1, PP-AO2) and one that contained 5 wt % of insoluble material (PP-AO3).

3.3. Oscillatory Shear Rheometry. As discussed above, oscillatory shear rheology provides insight into the architecture of the LCB chain population as well as the overall molecular weight of a PP derivative. Of particular interest are comparisons of η^* and G' between PP-AO and PP-VTES derivatives in the high-frequency domains that reflect the molecular weight and low-frequency domains that reflect the extent of branching. The data plotted in Figure 2 include the properties of the PP parent material, whose shear thinning and terminal flow characteristics are first described in Figure 1. Applying VTES formulations to this material resulted in a large-scale decline in molecular weight, with η^* falling by an order of magnitude in both LCB-PP derivatives. PP-S1 showed little evidence of branching, as the relationships between η^* , G' , and phase angle (δ) to oscillation frequency mirrored those of the linear starting material (Figure 2a). On the other hand, PP-S2 exhibited some evidence of an LCB architecture. While its complex viscosity at 100 rad/s was little different from that of PP-S1, clear evidence of melt elasticity was observed at low frequency, with δ approaching a plateau of less than 70°. These effects are attributed to the 2 wt % gel fraction of the sample,

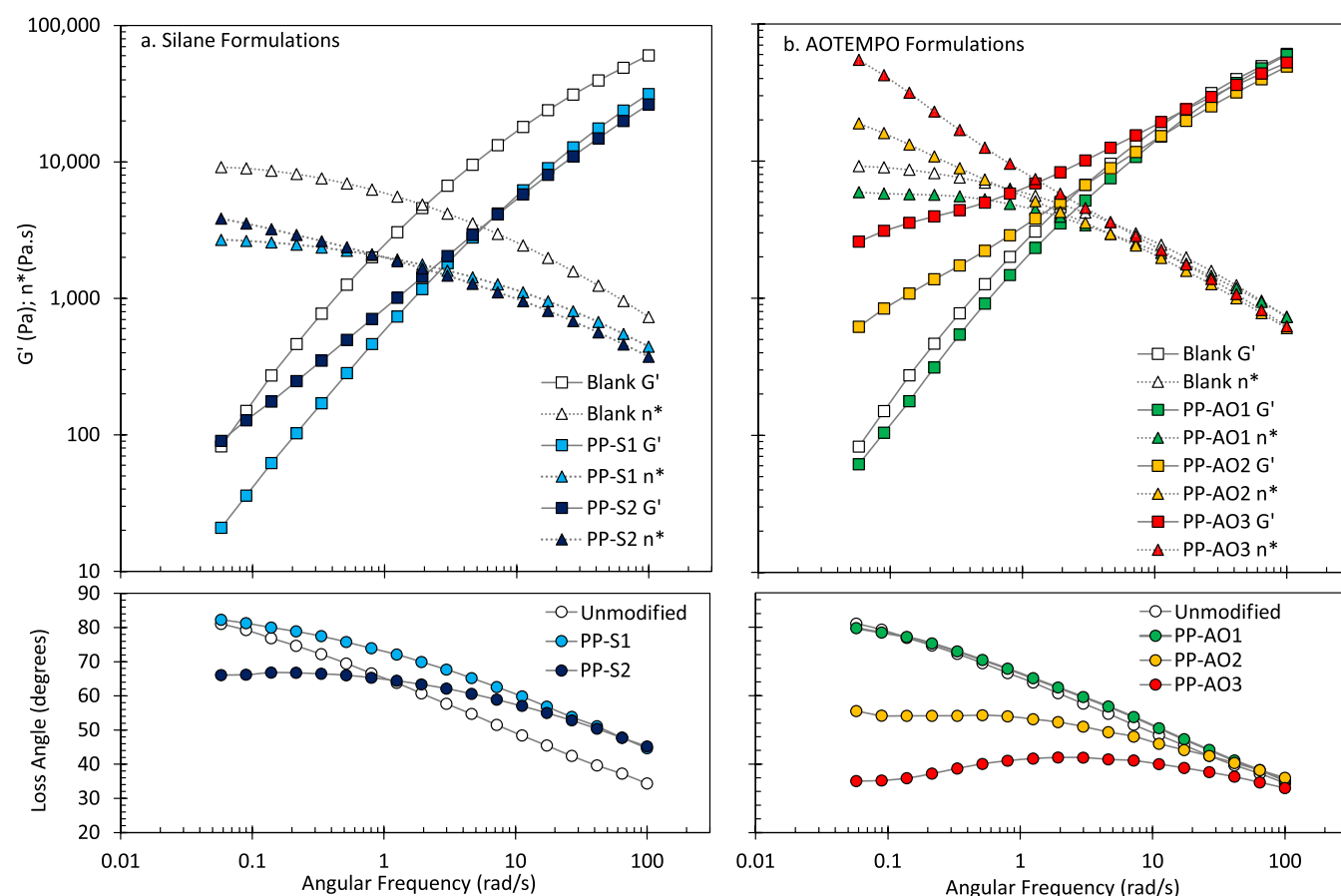


Figure 2. Storage modulus and complex viscosity of LCB-PP (a) silane formulations; (b) AOTEMPO formulations.

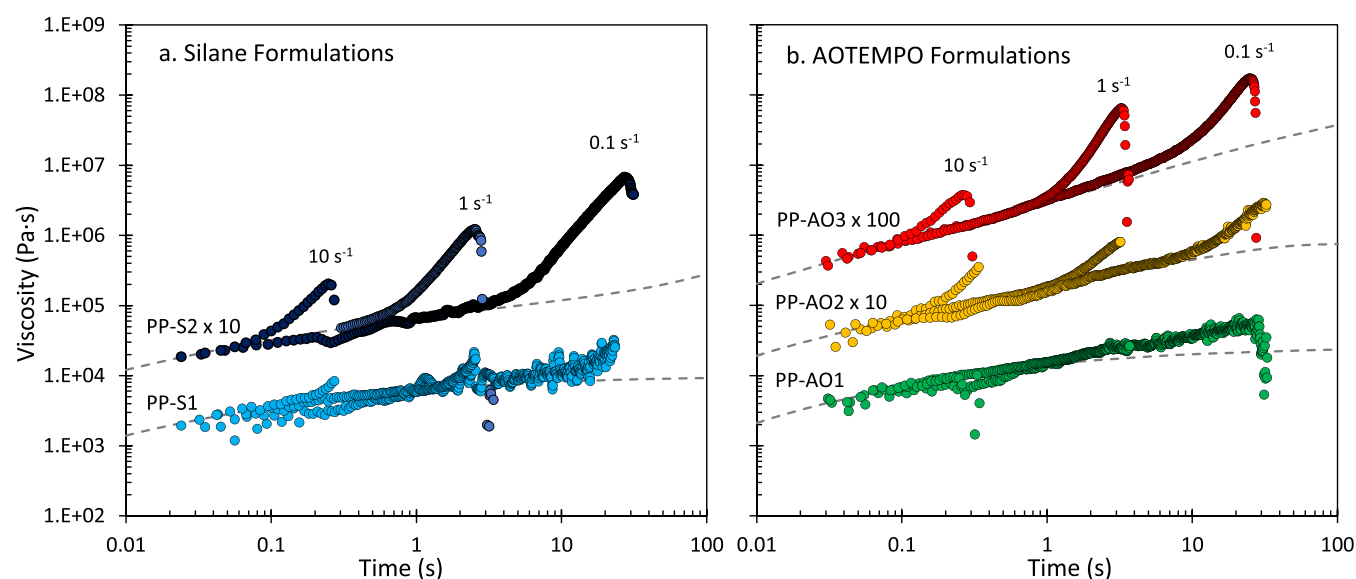


Figure 3. Extensional rheology of LCB-PP samples: (a) silane and (b) AOTEMPO. Dashed lines are $3\eta_0^+$.

whose slow relaxation over the timescale of the imposed deformation contributes significantly to melt-state properties.

The unique efficacy of AOTEMPO chemistry is revealed by the shear rheology data plotted in Figure 2b. All three derivatives maintained high-frequency properties while providing varying extents of low-frequency melt elasticity. With the least extent of polymer modification, the PP-AO1 formulation produced only subtle evidence of an impactful LCB chain

population, as its phase angle was nearly identical to that of unmodified PP. The PP-AO3 formulation imposed the highest extent of PP architecture modification, as the gel content eliminated the Newtonian plateau and elevated G' toward an equilibrium thermoset modulus.²³ PP-AO2 provided the most promising sample of this study, as this gel-free material demonstrated unambiguous evidence of long-chain branching along with clear indications of molecular weight preservation.

Moreover, the observed plateau in loss angle in the 0.1–1.0 rad/s range is a key feature of the time-dependent relaxation characteristics of an LCB material.²⁴ The high branch content of sample PP-AO3 is evidenced by much more pronounced low-frequency melt elasticity, as is expected for a polymer that contains an appreciable gel fraction.

3.4. Extensional Rheometry. Time-dependent elongational viscosities for all LCB-PP derivatives are plotted in Figure 3. Note that η_E^+ values for AOTEMPO-derived products were considerably higher than those produced by VTES formulations. For example, at a Hencky strain of 1 (below the strain-hardening threshold) and a strain rate of 1 s^{−1}, the extensional viscosity of PP-AO2 was 18 000 Pa·s, compared to 5800 Pa·s for PP-S1. This is attributed to molecular weight preservation by nitroxyl chemistry, which allows both shear and extensional viscosity to be predicted from the properties of the starting material.

Of the five LCB-PP samples prepared in this study, only PP-AO1 showed no evidence of strain hardening, with slight deviations of $\eta_E^+(t)$ from 3 times the time-dependent shear viscosity likely due to sample sagging effects at the low extension rate.²⁵ Strongly positive deviations from theoretical linear viscoelastic stress growth profiles were observed at high Hencky strains for PP-S2 and PP-AO3, as their strain-hardening response was bolstered by a substantial amount of infinite molecular weight gel. While this type of LCB-PP architecture may be acceptable for some applications, the stated goal of this work was to prepare gel-free materials with high melt viscosities along with extensional strain-hardening characteristics.

The silane-derived sample PP-S1 demonstrated significant strain hardening (Figure 3) despite exhibiting no evidence of branching in shear rheology—a testament to extensional rheology's greater sensitivity to LCB effects.²³ Of particular interest are the properties PP-AO2, which showed strong, progressive strain hardening at all extension rates, despite having no gel fraction.²⁶ Further insight into the branch architecture was gained by examining the strain-hardening coefficient, as defined below.²⁷

$$S = \frac{\eta_E^+(t, \dot{\epsilon}_0)}{3 \cdot \eta_0^+(t)}$$

The denominator is the time-dependent elongational viscosity as predicted by the Trouton relation, and the numerator is the measured elongational viscosity. In general, decreasing strain hardening with increasing strain rate is expected of materials with a small number of branches of high molecular weight.²⁸ The data plotted in Figure 4 represent a Hencky strain of 2 (lying above the onset of strain hardening, but below the ultimate strain for all trials). As expected, they varied with the extent of PP modification, with PP-AO3 producing the highest values. Moreover, the decline of *S* with increasing strain rate is consistent with a branching distribution comprised of both linear and LCB chain populations.²⁷

3.5. Crystallization Dynamics. The sensitivity of solid-state PP properties to semicrystalline morphology motivated us to examine the crystallization behavior of our LCB-PP derivatives.²⁹ Previous investigations have shown that crystallization from a melt state proceeds at a higher temperature (*T_c*) for branched polymers versus linear materials, but the observed *T_c* increase is relatively insensitive to the degree of branching.^{30,31} To determine whether our PP-AO products

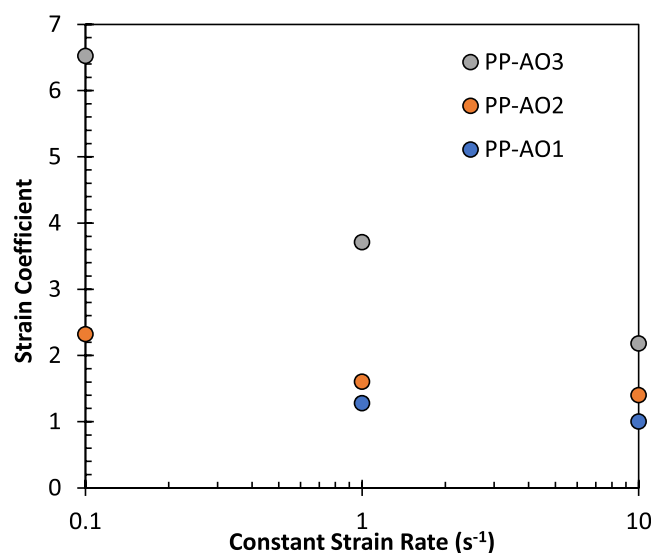


Figure 4. Strain-hardening coefficients for AOTEMPO-derived LCB-PP derivatives at a Hencky strain of 2.

conform to these general observations, nonisothermal DSC analyses at a scan rate of 5 °C/min were conducted to generate the crystallization exotherms shown in Figure 5. We note that,

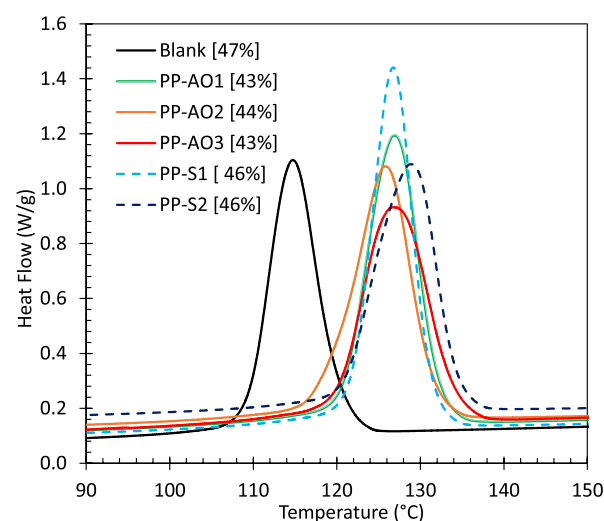


Figure 5. Nonisothermal crystallization exotherms of LCB-PP materials (degree of crystallinity in parentheses).

irrespective of the degree of PP functionalization and the chemistry employed, the *T_c* of LCB-PP was 10–14 °C greater than that of the linear parent material. Although studies have reported both increased^{29,32} and decreased^{33,34} crystallinity in branched polyolefins, variations in crystallization exotherms in the present work are considered to be within the error of the experimental technique.

The time-resolved isothermal DSC data plotted in Figure 6 reveal stark differences between the crystallization kinetics of LCB-PP and its linear parent material. For example, crystallization at 130 °C resulted in LCB-PP materials reaching their ultimate degree of crystallization within 5 min, compared to over 35 min for the blank linear precursor. Raising the crystallization temperature to 140 °C slowed the crystalline phase development for all PP materials while amplifying deviations between LCB-PP and PP and differentiating the

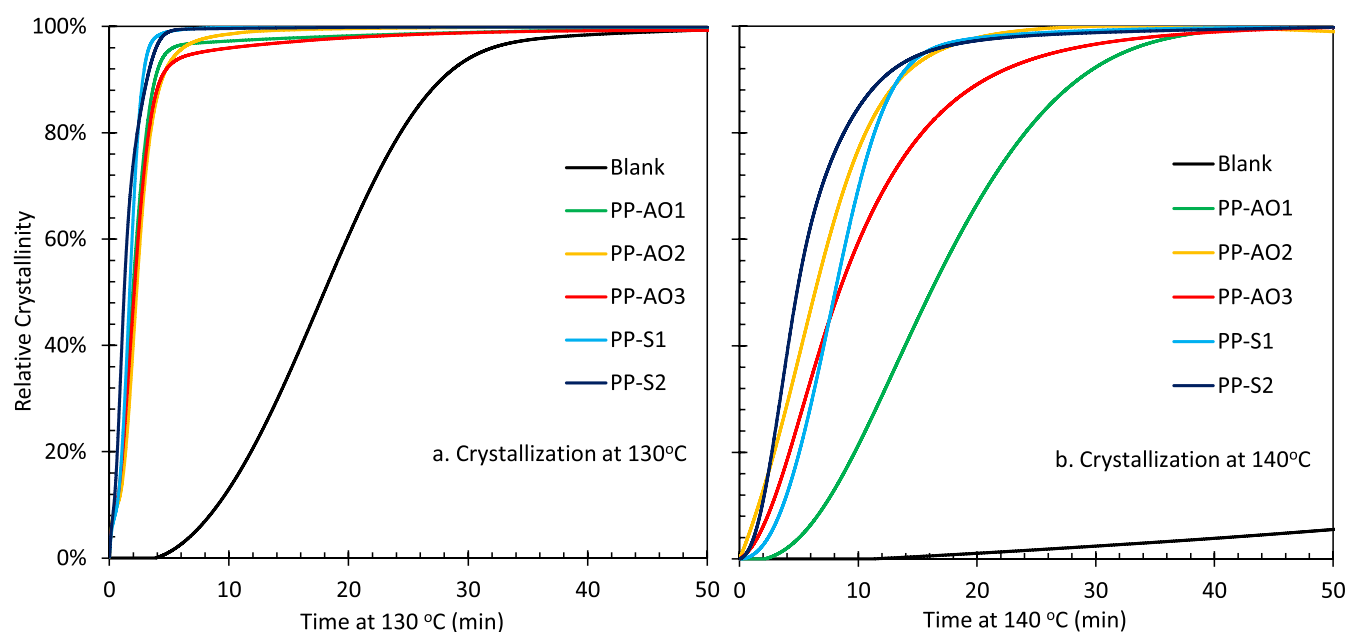


Figure 6. Isothermal crystallization of LCB-PP materials at (a) 130 °C and (b) 140 °C.

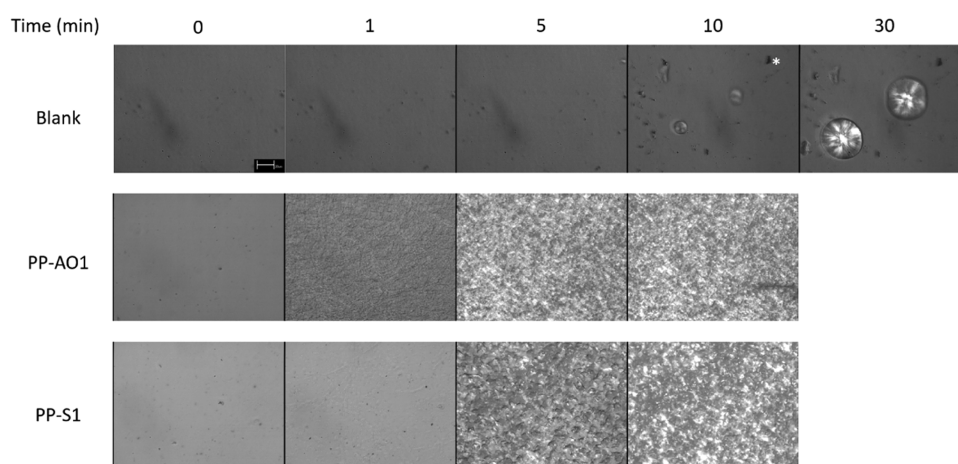


Figure 7. Polar optical microscopy of polymer crystallization at 140 °C. The scale bar is 20 μm .

lightly branched silane sample PP-S1 from the remainder of the field. We note that similar increases in isothermal crystallization rates were reported by Zhang et al. for silane-branched PP in their comparisons to coagent-based LCB materials.³³

The polarized optical microscopy images presented in Figure 7 complement DSC data by providing insight into the evolution of semicrystalline morphology throughout the crystallization process.^{29–31,35} They illustrate the genesis and growth of spherulites from the PP melt, with the ultimate morphology depending on the rate of crystallite nucleation in relation to the rate of spherulite growth. The data show that relative to linear PP, the gel-free AOTEMPO derivative PP-AO1 nucleates spherulites much more rapidly, resulting in accelerated crystallization and smaller spherulite size. This is consistent with recent reports on PP-LCB crystallization, which attribute this behavior to a nucleating effect of branch points.³⁵ Although definitive conclusions regarding the nucleation mechanism cannot be drawn from available information, the prodigious nucleation observed in these gel-

free branched samples supports the hypothesis that branched structures contribute to the nucleation of polypropylene crystals.

4. CONCLUSIONS

Peroxide-initiated formulations of AOTEMPO support single-stage, melt-state modifications of linear PP to yield gel-free LCB derivatives without incurring degradation of the polymer matrix. These PP-LCB products combine the high melt viscosity and extensional strain-hardening character that is required for many polymer processing operations. The introduction of an LCB architecture promotes the nucleation of crystallites from the melt, thereby increasing crystallization temperatures and generating finer spherulite morphologies.

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Notes

The authors declare no competing financial interest.

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